

Model $\rightarrow$ A parameterized Maths $f(x)$
whose parameters were Trained from data

Inference = running a trained model on new input to produce output.

Engine $\rightarrow$ execution of The model, such
That it runs, One Thing is To ~~me~~
Train it, after Training To run it
we run it on Engine.

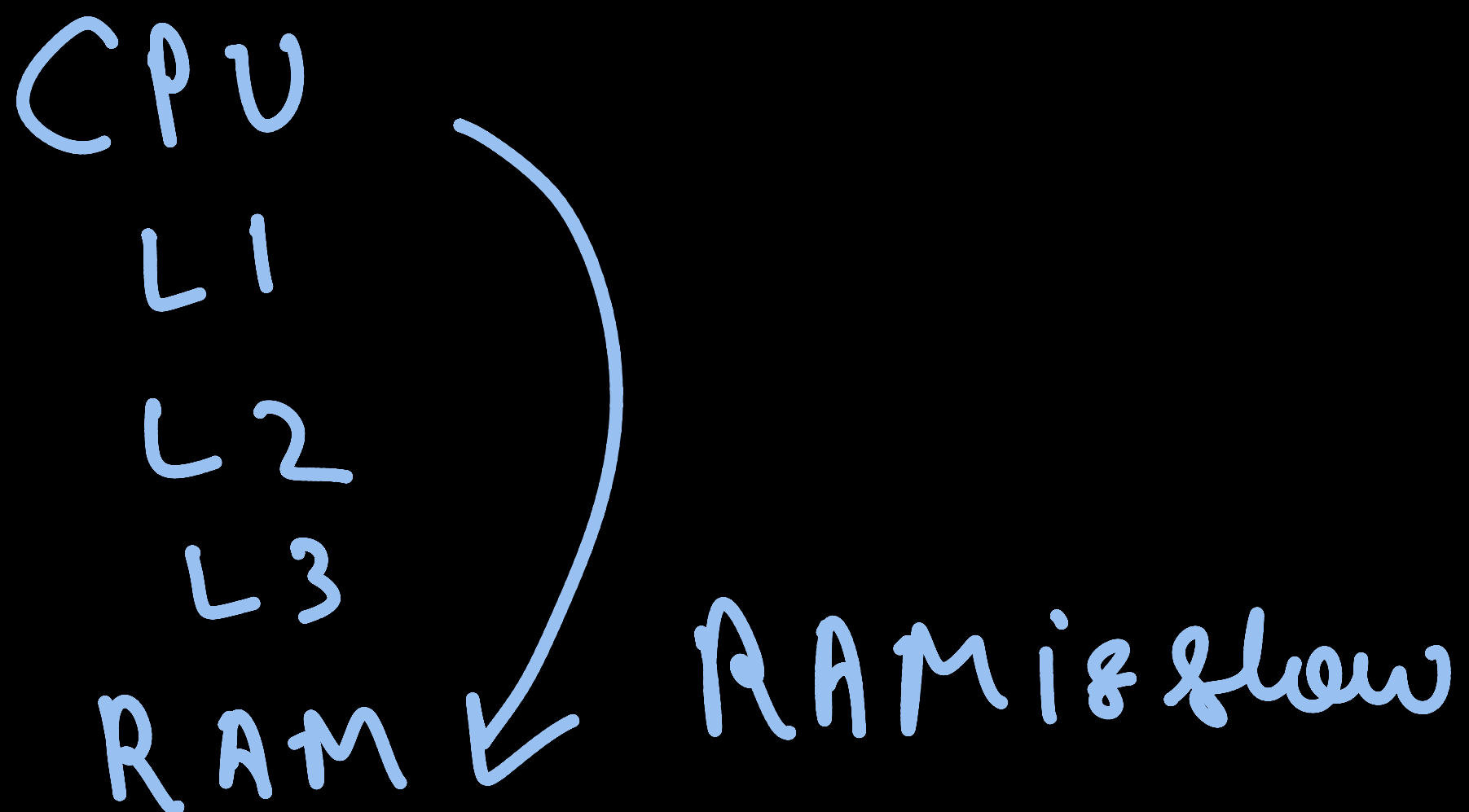
well the model now has now the brains, the ability to say i think this is a 7, and other would be the ability to run it, like getting input model runs model outputs kind of like humans, i see with eyes, eyes dont do any thinking brain does but brain needs eyes to see and then mouth just spits whatever i think again dont do thinking

Weight $\rightarrow$ numerical Strength value
showing significance.

Neuron $\rightarrow$ Receives number
Multiply by weights
Sum it

Sum 12
Apply activation on it.

CACHE → To avoid Slow memory access



Under:

```
<> Python  
model(x)
```

there might be:

```
Python  
↓  
Framework runtime  
↓  
Tensor operations  
↓  
Optimized kernels  
↓  
CPU/GPU instructions  
↓  
Hardware
```

That's the hidden systems stack.

Matrix Multiplication To avoid

Over and over dot product.

- **A model**
=
a mathematical structure containing learned numerical parameters.

Weights

=
numbers controlling how strongly inputs affect outputs.

A neuron

=
weighted sum + activation.

A layer

=
many neurons together.

Matrix multiplication

=
efficient way to compute many neurons at once.

Activation provides non-linearity in Model

- if weight is different then model is different

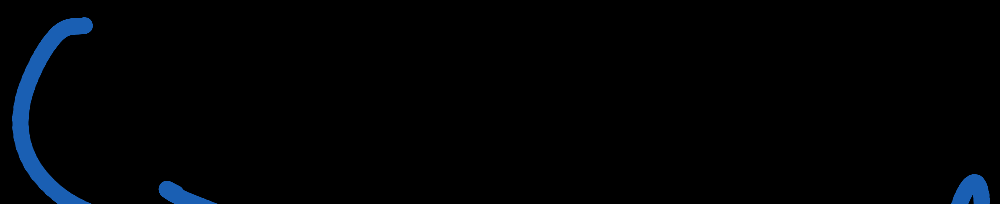
Ex : 28×28 grid images

= 784 number



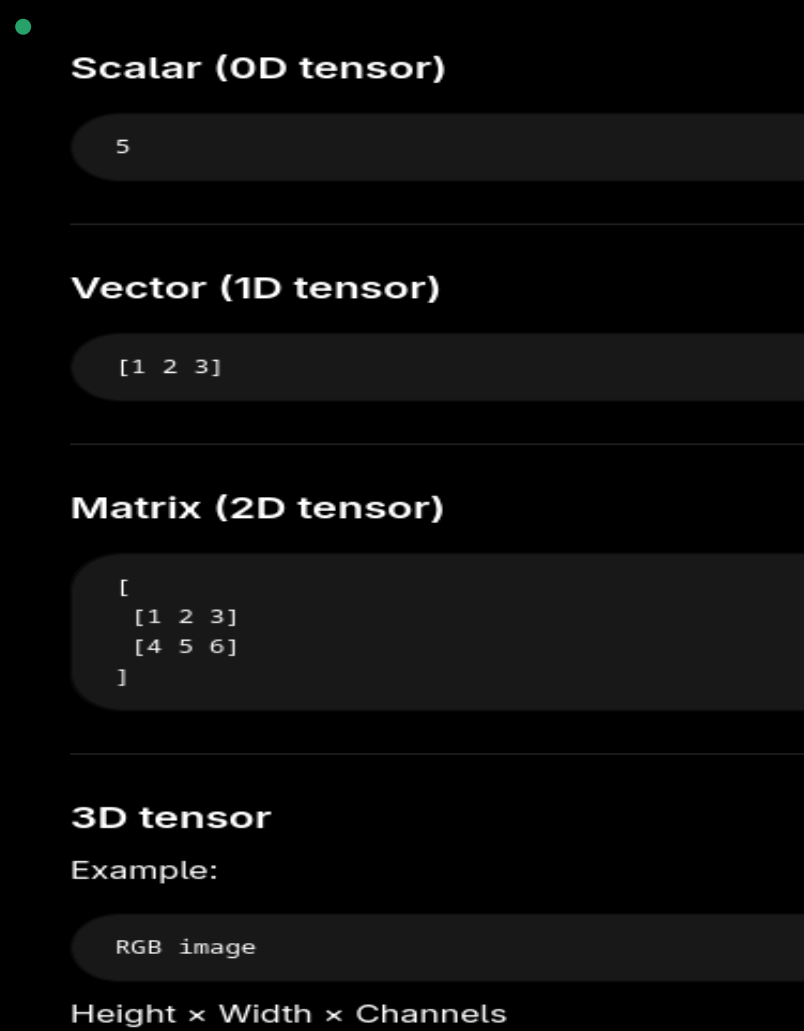
128 neurons

Can be any
power of
2



→ no output classes.

Tensor: multidimensional numerical array.



Where does for Row major

$$\text{address} = i \times \text{Col} + j$$

Come from

ex	a	b	c
	d	e	f

in memory

a b c d e f

To get to [1][1]

↙

Cross 1 row

To cross

and add
offset.

$$\underline{i \times \text{Col} + J}$$

offset

1 2 3 4

← cross

5 6 7 8

← There

9 10 11 12

reach here

Then add

offset

offset addition.

1 2 3 4 5 6 7 8 9 10 11 12